Internet Programming (Sessional) [CSE–326]

Proposal Report on:

Academic Radar

Group No: 10

Name	Student ID
Mahmudul Islam	2204027
Shrabon Barua	2204003
Abdullah Al Azim	2204012

Submitted to:

Md. Rashadur Rahman
Assistant Professor
Department of CSE, CUET

Department of Computer Science and Engineering (CSE)
Chittagong University of Engineering & Technology (CUET)
Chattogram–4349, Bangladesh

0.1 Objectives

The key objectives of this project are:

1. To detect students who are academically at risk at an early stage using measurable academic indicators.
2. To support timely advisor–student interaction by providing advisors with early academic risk insights.
3. To enable preventive academic guidance before failure or dropout occurs.

0.2 Introduction

In many universities, academic problems are usually noticed only after a student has already failed or fallen far behind. However, academic failure rarely happens suddenly. It often develops over time due to factors such as poor attendance, declining grades, missed class tests, assignments and reduced engagement in academic activities. Because these warning signs are not tracked systematically, advisors often become aware of the situation too late to provide effective support.

Academic Radar is proposed as a preventive academic support platform that focuses on early detection rather than post-failure analysis. The system is designed to assist academic advisors by identifying students who are at risk at an early stage and encouraging timely advisor–student communication before academic failure or dropout occurs.

0.2.1 Motivation

Academic advisors play an important role in guiding students, but they often have limited access to real-time academic progress data. Many students who struggle academically hesitate to seek help on their own, while advisors typically receive information only after official evaluations or results are published. This delay reduces the effectiveness of academic advising.

The main motivation of this project is to reduce this gap by introducing an early-warning system that improves advisor awareness. By providing timely insights into academic risk, the system enables supportive interventions without penalizing or stigmatizing students.

0.3 Potential Stakeholders

1. **Students:** The primary users who can privately view their academic risk status and receive guidance to improve their performance.
2. **Advisors / Faculty Members:** Can receive early alerts, review risk indicators, and take appropriate supportive actions.
3. **Academic Administration:** Can analyze overall academic risk trends to support academic planning and policy decisions.

0.4 Existing Related Work

1. **Development of an early warning system for higher education institutions by predicting first-year student academic performance:**
   https://www.researchgate.net/publication/382275896_Development_of_an_early_warning_system_for_higher_education_institutions_by_predicting_first-year_student_academic_performance

2. **Awareness to Action: Student Knowledge of and Responses to an Early Alert System:**
   https://www.mdpi.com/2076-3417/15/11/6316

3. **Predicting student dropouts with machine learning: An empirical study in Finnish higher education:**
   https://www.sciencedirect.com/science/article/pii/S0160791X24000228

0.5 Contribution of the Work

1. Design and implementation of a rule-based academic risk prediction system using attendance and performance data.

2. Introduction of an advisor-focused alert mechanism to support timely academic intervention.

0.6 Proposed Features of the Web Application

1. **Student Data Collection Module:** Collects attendance records, assessment scores, and GPA trends.

2. **Risk Scoring Engine:** Computes academic risk scores based on predefined rules and thresholds.

3. **Pattern Detection Module:** Detects gradual academic decline as well as sudden drops in performance.

4. **Advisor Alert System:** Notifies advisors when a student reaches a critical risk level.

5. **Intervention Tracking Module:** Records advisor actions and monitors outcomes after intervention.

6. **Analytics Dashboard:** Presents summarized academic risk data for administrative analysis.

0.7 Expected UI Design

1. Login Page

```
                    Academic Radar Login

Email          : []
Password       : []

 Select Role   : ( ) Student   ( ) Advisor ( ) Admin

[ Login ]           Sign Up      [Forgot Password]
```

2. Student Sign Up

```
                      Student Sign Up

Full Name        : []   Roll No : []
Batch            : []   Current Semester : []
Hall Name        : []
Mobile No        : []   Email : []
Address          : []
Password         : []   Confirm Password : []

 [ Create Account ]              [ Back to Login ]
```

3. Student Dashboard

```
                      Student Dashboard

Name            : Rahim Ahmed
 Student ID      : 2204145
 Level-Term      : Level 3 - Term 1
 Completed Semesters : 5 / 8

 Risk Level      :  WATCHLIST
 Academic Stage : Gradual Decline

 Average Attendance : 71%
 Semester GPA (SGPA)  : 2.94
 Overall CGPA         : 3.01 ↓ (from 3.36)

 [ Enter / Update Academic Data ]
 [ View Detailed Results ]
 [ Messages from Advisor ]
```

4. Academic Data Entry (Student)

```
Academic Performance Submission

Semester        : Level 2 - Term 1

Subject: Data Structures (Non-Sessional)
Credit: 3.0
Attendance (%) : []
Class Tests (n+1 = 4)
CT1 : []  CT2 : []  CT3 : []  CT4 : []
Final Exam : [] / []

Subject: DBMS Lab (Sessional)
Credit: 1.5
Attendance (%) : []
Final Quiz : [] / []
Final Viva : [] / []
Continuous Assessment :
  Report : [] / []
  Daily + Final Performance : [] / []

[ Submit Data ]
```

5. Subject-wise Result (Student View)

```
Subject Result Summary

Subject : Data Structures
Credit  : 3.0
Total Obtained CGPA: A
Percentage     : 74%
Grade          : A-
Grade Point (GPA) : 3.50
```

6. Semester CGPA & Overall CGPA (Student View)

```
Semester & CGPA Summary

Semester : Level 2 - Term 1

Subject              Credit     CGPA
Data Structures       3.0      3.50
Algorithms            3.0      2.75
DBMS Lab              1.5      3.25
Math IV               3.0      2.50

Semester CGPA (SGPA) : 2.94
Overall CGPA        : 3.01 ↓ (from 3.36)
Risk Level          : WATCHLIST
```

7. Advisor Dashboard

```
                    Advisor Dashboard

 Advisor : Dr. S. Karim
 Department : CSE
 Assigned Students : 16

 Risk Overview:
  High Risk : 3
  Watchlist : 6
  Safe : 7

 Students (Sorted by Risk):
 ID            Name        CGPA    Risk
 2021-CSE-032  Anik      2.21   □ High
 2021-CSE-045  Rahim      3.01   □ Watch
 2021-CSE-061  Hasan      3.54   □ Safe

[ Select Student for Details ]
```

8.Student Detail (Advisor View)

```
              Student Performance Analysis

 Name        : Rahim Ahmed
 Level-Term : L2 - T1
 Attendance Avg : 71%
 SGPA            : 2.94
 CGPA            : 3.01 ↓ (from 3.36)
 Risk Level      : □ WATCHLIST
 Pattern        : Gradual Academic Decline
 Risk Type      : Attendance-Dominant Risk

 Advisor Actions:
[ Send Message ] [ Schedule Counseling ]
[ Issue Academic Warning ] [ Log Intervention ]
```

9.Advisor Notification to Student

```
                Advisor → Student Message

Student :

Message :

[ Send Notification ]
```

10. Intervention Tracking (Advisor)

```
                    Intervention Log

 Student : Rahim Ahmed
 Advisor : Dr. S. Karim

  Action Taken:
  Counseling Session
  Advisor Remarks:
  Date :

 [ Save Record ]
```

11. Admin Dashboard

```
               Admin Analytics Dashboard

 Academic Year :
 Total Students :

 Risk Distribution :
  High Risk : 15%
  Watchlist : 28%
  Safe : 57%

Department-wise High Risk :
 CSE : 18%  EEE : 21%  BBA : 12%

 Policy Insights :
• Attendance below 75% is main risk factor

Trend Analysis : Dropout Risk : Decreasing
[ View Detailed Reports ]
```